PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

Team: T-Junction

Project Name: Zen Travel Your AI Travel Concierge

Domain: AI Systems & Agentic Workflow Automation

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1. Project Overview

Problem Statement: Despite the increased accessibility of global travel, the process of planning a trip has become a significant psychological and financial burden for the modern consumer. Travelers today face three primary barriers:

- **Economic Pressure:** Skyrocketing inflation and stagnant wages have transformed holidays from a standard yearly expectation into a high-stakes financial luxury.
- **Information Overload:** The "Paradox of Choice" in the digital age. A typical traveler spends hours navigating fragmented platforms, comparing volatile pricing, and vetting the authenticity of reviews and third-party booking sites. This leads to "decision fatigue" before the trip begins.
- **The Autonomy Gap:** Current alternatives fail to bridge the gap between effort and enjoyment. Independent planning is high-effort and stressful, while group tours strip autonomy with rigid schedules, "kickback" shopping stops, and zero personalisation.

Target Domain: AI Systems & Agentic Workflow Automation applied to end-to-end travel planning, from discovery and itinerary generation to visa, arrival cards, and on-trip assistance.

Proposed Solution Summary: Zen Travel is an agentic travel concierge powered by Z.AI's GLM as the central reasoning engine. A traveller speaks or types in any of 11 supported languages, and the GLM-driven orchestrator decomposes the request into a multi-step workflow: preference profiling, destination shortlisting, budget-aware itinerary generation, visa and Malaysia Digital Arrival Card (MDAC) preparation, and live in-trip Q&A. Each step calls the right tool (search, maps, PDF generation, Supabase persistence, face-liveness auth) and produces a structured, actionable output collapsing days of fragmented planning into a single guided conversation.

2. Background & Business Objective

Background of the Problem: Travel planning today is spread across 10+ disconnected surfaces flight aggregators, hotel sites, blogs, review platforms, embassy portals, currency converters, and

messaging apps. Each surface has its own format, language assumptions, and biases. Older travellers and non-English speakers are systematically excluded; younger travellers burn out from comparison fatigue. Group tours solved the effort problem but introduced a new one: travellers lose autonomy and authenticity.

Importance of Solving This Issue: Travel is one of the largest discretionary-spend categories globally and a meaningful share of Malaysia's GDP. Reducing planning friction unlocks demand from segments that currently opt out budget-conscious families, elderly travellers, and first-time international visitors while raising trip quality for everyone else. An AI workflow that handles ambiguity, missing data, and process failures (visa rejections, sold-out stays, schedule clashes) directly addresses the domain's core expectations.

Strategic Fit / Impact: Zen Travel aligns with the AI Systems & Agentic Workflow Automation domain by using Z.AI's GLM as the irreplaceable reasoning core: remove the GLM and the system can no longer interpret unstructured traveller intent, sequence the planning steps, or adapt when a step fails. The platform also supports Malaysia's tourism objectives by automating MDAC submission and surfacing local, lesser-known destinations alongside headline ones.

3. Product Purpose

3.1 Main Goal of the System: To deliver an AI-driven, end-to-end travel workflow that lets any traveller regardless of language, age, or planning experience go from a vague intent ("a calm 5-day trip under RM3,000") to a booked, documented, and supported journey without leaving the platform.

3.2 Intended Users (Target Audience):

- Gen Z and millennial independent travellers seeking authentic, budget-aware itineraries
- Elderly and first-time international travellers who need guidance in their native language
- Families coordinating multi-person trips with mixed preferences and budgets
- Non-English speakers across 11 supported languages (EN, BM, ZH, TA, JA, KO, AR, FR, ES, DE, HI)

4. System Functionalities

4.1 Description: Zen Travel operates as a stateful, multi-agent workflow engine. The user's natural-language input enters an Intent Router, which the GLM uses to classify the request (plan, modify, document, support) and dispatch it to the relevant sub-workflow. Each sub-workflow is a chain of GLM reasoning steps interleaved with tool calls (search, maps, PDF, Supabase, payment mocks). Workflow state preferences, draft itinerary, documents is persisted per user, so the conversation can resume across sessions.

4.2 Key Functionalities:

- **Agentic Trip Planning:** Conversational onboarding captures traveller persona, budget band, pace, dietary needs, and accessibility constraints in 4 guided steps; output is a structured profile stored in Supabase.
- **Itinerary Synthesis:** GLM generates a day-by-day itinerary balancing must-see anchors and hidden-gem detours, respecting opening hours, travel time, and the budget envelope. Output is rendered on an interactive Leaflet map and a draggable timeline.
- **Multilingual AI Concierge:** A floating concierge accepts text or voice input, answers in the user's language, and can mutate the itinerary in place ("swap day 2 for something quieter").
- **Visa & MDAC Automation:** GLM identifies the visa regime for the traveller's nationality + destination pair, drafts the MDAC submission payload, and exports a pre-filled PDF via pdf-lib.
- **Face-Liveness Auth:** AWS Rekognition + MediaPipe face-liveness gate the user account, ensuring booked documents map to a real, present human not a recycled credential.
- **Stateful Recovery:** Workflow state is checkpointed after each GLM call so a failed external API (e.g. map service down) does not lose the draft itinerary.

4.3 AI Model & Prompt Design

4.3.1 Model Selection: ILMU GLM 5.1 is the central reasoning engine. GLM was chosen for its strong multilingual coverage (critical for our 11-language requirement), competitive tool-use / function-calling reliability, and cost profile that makes per-session agentic workflows financially viable. Removing GLM collapses the product: no other component can decompose ambiguous travel intent or coordinate the downstream tools.

4.3.2 Prompting Strategy: Multi-step agentic prompting with role-scoped sub-prompts. A top-level Planner prompt performs zero-shot intent classification and emits a JSON workflow plan. Each step is then executed by a specialised prompt (Profiler, Itinerary-Builder, Visa-Advisor, MDAC-Filler, Concierge) that receives only the slice of state it needs. Few-shot examples are injected into the Itinerary-Builder and MDAC-Filler prompts where format fidelity matters; everything else stays zero-shot to control token cost.

4.3.3 Context & Input Handling: Maximum accepted user input is 8,000 characters per turn. Conversation history is summarised into a rolling profile object once it exceeds 4,000 tokens, so we never ship raw history past that bound. Uploaded documents (passport scans, prior bookings) are OCR'd, then chunked at 1,500 tokens with 150-token overlap; chunks are embedded and only the top-3 relevant chunks are injected into the active prompt. Inputs above the cap are rejected with a localized error that asks the user to shorten or split.

4.3.4 Fallback & Failure Behavior: Every GLM call returns JSON validated against a per-step schema. On schema failure the system retries once with a stricter "return JSON only" reminder; a second failure triggers a graceful fallback that surfaces a partial result plus a human-readable explanation. Hallucinated facts (e.g. non-existent attractions) are caught by a post-generation grounding check against the maps/search tool unverified entities are flagged in the UI, not silently shown. Sustained failures route the user to a human-handoff form that captures the conversation transcript.

5. User Stories & Use Cases

User Stories:

- "As a first-time solo traveller, I want to describe my trip in my own words and get a complete day-by-day plan, so I do not have to compare 20 tabs."
- "As a Mandarin-speaking grandmother, I want to use the app entirely in Chinese with voice, so language is never the reason I cannot travel alone."
- "As a budget-conscious family planner, I want the AI to keep total spend under my cap and warn me before any change blows the budget."
- "As a tourist arriving in Malaysia, I want my MDAC pre-filled and downloadable as a PDF, so I clear immigration without panicking."

Use Cases (Main Interactions):

- **Onboarding & Profile Capture:** User logs in via face-liveness, completes 4-step onboarding; GLM produces a structured traveller profile and seeds the recommendation engine.
- **Plan-from-Prompt:** User types "5 days in Penang under RM2,500, calm pace, vegetarian." GLM emits a workflow plan, calls map + search tools, then assembles a draft itinerary rendered on the timeline + map view.
- **In-Trip Conversational Edit:** User opens the floating concierge mid-trip and asks (in any of 11 languages) for a replacement for "tomorrow afternoon too hot." The agent rewrites only that slot, preserving the rest.
- **Visa + MDAC Automation:** User uploads passport; GLM extracts nationality, looks up visa requirements for the destination, drafts the MDAC payload, and exports a pre-filled PDF.

6. Features Included (Scope Definition)

- Conversational onboarding with a structured 4-step profile capture
- GLM-driven multi-step itinerary generation with map + timeline rendering
- Multilingual concierge supporting 11 languages, text and voice input
- Visa requirement lookup and MDAC pre-fill + PDF export
- Face-liveness authentication via AWS Rekognition + MediaPipe
- Persistent per-user workflow state in Supabase with session resume
- Explore feed surfacing budget-aware destinations and hidden gems
- Schema-validated GLM outputs with retry, grounding, and graceful fallback

7. Features Not Included (Scope Control)

- Real flight, hotel, or activity booking payment is mocked; we do not integrate live GDS or OTA inventory in this version
- Group / multi-user simultaneous editing of the same itinerary
- Native mobile apps the platform is a responsive web app only for the hackathon
- Offline mode the agentic workflow assumes network connectivity to GLM and external tools

- Insurance brokering, currency hedging, and post-trip expense reconciliation

8. Assumptions & Constraints

LLM Cost Constraint: Estimated 12k15k GLM tokens per average planning session (profile + itinerary + 35 concierge turns). To keep inference economical at scale we (a) cache deterministic sub-prompts (visa rules per nationality+destination pair, language-specific system prompts) keyed by hash, (b) summarise history past 4,000 tokens, and (c) route low-complexity follow-ups ("what's the weather") to a smaller GLM tier.

Technical Constraints: Frontend is Next.js 15 / React 19 / Tailwind; backend is FastAPI (Python 3.10+). Face liveness depends on AWS Rekognition availability in the deployed region. PDF export is limited to MDAC and itinerary; arbitrary government forms outside MDAC are out of scope.

Performance Constraints: Itinerary generation targets under 12 seconds end-to-end on a warm cache; cold starts may take up to 25 seconds because of tool-call fan-out. Concierge turns target under 4 seconds. The GLM provider's per-minute rate limit caps concurrent active sessions.

User Input: The system requires explicit traveller intent it does not auto-plan trips without a user prompt. Final bookings (when integrated post-hackathon) will require human confirmation; the agent never commits a payment autonomously.

9. Risks & Questions Throughout Development

Hallucinated Recommendations: How do we guarantee that every attraction, restaurant, or hotel the GLM suggests actually exists and is currently operating? Our mitigation is post-generation grounding against the maps/search tool, but edge cases (recently closed venues) remain.

Multilingual Quality Drift: Does itinerary quality stay consistent across all 11 languages, or do low-resource languages (e.g. Tamil, Arabic) get degraded reasoning? We plan a per-language eval set and a fallback that drafts in English then translates if quality dips.

Workflow Recovery on Tool Failure: If the maps API or PDF service fails mid-workflow, can we resume from the last checkpoint without forcing the user to restart? Current design checkpoints after every GLM call we still need to validate this end-to-end under chaos testing.

Privacy of Sensitive Documents: Passport scans and MDAC data are sensitive. How long do we retain them, where, and under what user consent? Default plan: encrypted at rest in Supabase, auto-purged 30 days post-trip, with explicit opt-in for retention.

Cost at Scale: If GLM token cost rises or session length grows, can the unit economics still work? We will track tokens-per-completed-itinerary as a north-star efficiency metric.